JESUS ANDRES RUIZ

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EDUCATION

University of Texas at San Antonio **UTSA Honors College | GPA: 3.43**
Bachelor of Science in Mechanical Engineering *Expected May 2028*

PROJECTS

Exoskeleton Hip Module **May 2026 – Present**
Undergraduate Research Assistant

- Led failure analysis of cracked cuffs, sliders, and struts on early prototypes; redesigned 4 components in Fusion 360 and validated each with static stress simulation.
- Iterated the hip abductor through 3 revisions to a machined, anodized 6061-T6 aluminum part, cutting peak von Mises stress 20% and widening internal fillets from 1.0 to 1.6 mm for CNC tooling.
- Fabricated parts via Fused Deposition Modeling, carbon fiber printing, CNC machining, and outsourced cutting; caught a 200 µm bore deviation from anodizing buildup before final assembly.

Bowden Cable Transmission — Ankle Actuation **May 2026 – Present**
Undergraduate Research Assistant

- Prototyped a Bowden cable transmission with graduate researchers that moves the 485 g ankle actuator off the distal leg to reduce swing-leg inertia.
- Designed, printed, and tested 2 TPU cable insert geometries (tapered vs. straight), selecting the tapered design for better cable routing and fit.

UTSA Formula Society of Automotive Engineers **August 2025 – May 2026**
Team Member — Chassis / Brakes

- Designed 4 aluminum caliper spacers in SolidWorks with GD&T thickness and parallelism callouts, then turned and faced them on a lathe to keep pad contact even and rotor clearance consistent.
- Collaborated with a 5-person chassis and brakes sub-team to iterate designs under FSAE rules, supporting fitment checks and vehicle assembly.

EXPERIENCE

UTSA Department of Mechanical Engineering **August 2026 – Present**
Teaching Assistant, Engineering Practice and Graphics

- Hold office hours weekly, helping students hand-sketch orthographic, isometric, and section-view drawings with correct line conventions and dimensioning.
- Grade homework and exams, providing written feedback on drawing accuracy, projection, and dimensioning errors.

UTSA Intelligent Systems and Bionics Engineering Lab — Lab Role & Mentorship **May 2026 – Present**
Undergraduate Research Assistant

- Fabricated and assembled exoskeleton hardware alongside graduate researchers, producing printed and machined parts across multiple lab projects (see Exoskeleton Hip Module and Bowden Cable Transmission above).
- Delivered weekly research progress updates to Dr. Binghan He, documenting design changes, test results, and next steps.

UTSA Hypersonics Lab **January 2026 – May 2026**
Undergraduate Research Volunteer

- Milled 50+ precision diaphragms for a Mach 7.2 Ludwieg tube, setting burst pressures of 200–400 psi through material thickness and scored groove depth.
- Supported graduate researchers across 30+ wind tunnel runs, aligning laser-based Schlieren imaging to visualize shock waves and prepare instrumentation for data collection.

SKILLS

Software & Tools: SolidWorks, Fusion 360 (CAD + Simulation), MATLAB, Microsoft Office
Programming / Embedded: Arduino, Python, Git/GitHub
Manufacturing & Design: Geometric Dimensioning and Tolerancing, Rapid Prototyping, Root Cause Analysis
Fabrication & Tools: Fused Deposition Modeling & Continuous Carbon Fiber Printing, Soldering, Welding, CNC Machining
Coursework: Statics, Dynamics, Thermodynamics I, Materials Engineering & Lab, Engineering Practice & Graphics